

Titanic EDA – Findings Report

Titanic Dataset — Exploratory Data Analysis (EDA)

This notebook performs a thorough EDA on the Titanic training dataset. It includes:

- Data loading & quick look
- Data cleaning & missing value analysis
- Univariate analysis (numerical & categorical)
- Bivariate analysis focusing on `Survived`
- Correlation analysis & heatmap
- Feature engineering suggestions and visualizations

Dataset path: `train.csv`

Summary of findings & next steps

- **Missing values:** Age and Cabin have the most missing values. Age can be imputed (by Title or median); Cabin may be condensed to deck or use 'has_cabin' flag.
- **Key predictors of survival:** Sex (female higher), Pclass (lower class less survival), Fare (higher fares more likely to survive), Title (e.g., Mrs/Miss/Master), Age has a nuanced effect.
- **Feature engineering suggestions:** FamilySize, Title, HasCabin, Deck extracted from Cabin, Age bins, Fare bins.
- **Next steps:** Clean Cabin, impute Age, transform skewed Fare (log), scale features, then train & validate predictive models (logistic regression, random forest) and evaluate.